

1. What do you understand by crop-water requirement? Explain various factors affecting the crop-water requirement.

Answer:

Crop Water Requirement (CWR) is defined as the total quantity of water required by a crop from the time it is sown to the time it is harvested. The Mahabharat region of Nepal extends from the altitude of 1500-3300 m above the mean sea level, has a subtropical to temperate climate, with most rainfall during the monsoon (June–September). Hence, Supplementary irrigation has to be provided, especially during the dry season.

The major factors affecting crop-water requirement are explained below:

i. Climate:

- Higher temperature increases evaporation and transpiration increasing the crop-water requirement.
- Effective rainfall reduces the amount of irrigation required.
- Greater sunshine increases evapotranspiration and crop-water requirement.
- String winds increase evaporation and transpiration resulting in greater water requirement.

ii. Type of crop:

- Different crops require different amount of water for their growth.
- Crops such as rice and sugarcane require more water.
- Crops such as maize, wheat, pulses require comparatively lesser water.

iii. Soil characteristics:

- Crop water requirement also varies depending upon the type and properties of soil.
- Sandy soils have high infiltration and low water-holding capacity so they require more frequent irrigation.
- Clayey soils have high water holding capacity and requires less frequent irrigation.

iv. Method of irrigation:

- Amount of water to be supplied also depends upon the efficiency of the irrigation method.
- Flood irrigation requires more water due to conveyance and application losses.
- Sprinkler irrigation reduces some losses.
- Drip irrigation supplies water directly to the root zone and requires less water.

v. Cropping pattern:

- The type of crops, sequence and combination in an area also affect the total water requirement.
- A cropping pattern containing water intensive crops requires more irrigation water.

2. Name various type of irrigation methods. Describe in brief the advantages and disadvantages of each method.

Answer:

Irrigation is an artificial application of water to agricultural land in the required quantity and at the required time to supplement natural rainfall for optimum crop growth and yield.

There are several methods of irrigation adopted based on the suitability of local condition such as soil characteristics, topographic conditions, crop rotation, available water flow and others. Few methods of irrigation are briefly described along with its advantages, limitations and suitability.

i. **Surface irrigation:**

- Surface irrigation is the most common type of the irrigation where the water is applied on the field in varied quantities and at different times.

- In this method, stream of water is diverted from the head of a field into the furrows or borders and allowing it to flow downward.
- Advantages:
 - a. It allows to store the required amount of water in the capillary zone of the soil for supply of root zone of plants.
 - b. It allows the use of machinery for land preparation cultivation and harvesting.
- Limitations:
 - a. Greater loss of water by surface runoff and deep percolation.
 - b. Inferior quality crops with a low yield are produced from the surface irrigation.
- Suitability:
 - Surface irrigation is most suitable for the land with the gentle slopes like terai region of Nepal.
 - This method is suitable for row and field crops.



ii. Sprinkler irrigation:

- Sprinkler irrigation is an improvement over conventional surface irrigation method.
- This method simulates natural rainfall to spread water in the form of rain uniformly over the land surface as needed at uniform pattern and at less rate than the infiltration rate of the soil to avoid the surface runoff.
- Advantages:
 - Provides protection to small plants from wind damage and soil from blowing.
 - Sprinklers provides protection against freezing injury to crops.
- Disadvantages:
 - Initial cost of installation is relatively high and regular maintenance must be done due to frequent clogging of the system.
 - This method is not so suitable at plain land as pressure head and sediment free water is required.
- Suitability:
 - Sprinkler irrigation is highly suitable for sandy soils, rolling and uneven terrain.
 - It is suitable for closely spaced crops like cereals and fodders.



iii. Drip irrigation:

- Drip irrigation is the method of applying water directly to the plants through a number of low flow rate outlet called emitter or drippers, placed at short interval along with small tubing.
- The network of drippers supply water and fertilizer to the plant roots at the rate of 1 to 2 liter per hour.
- **Advantages:**
 - Reduced percolation and evaporation losses resulting in economical use of water.
 - Fertilizer and nutrient losses minimized due to localized application and reduced leaching.
- **Disadvantages:**
 - Frequent clogging of the dripper requires constant supervision.
 - Not suitable for very closely seeded crops such as carrots, radishes etc.
- **Suitability:**
 - This method of irrigation is highly suitable for arid regions and sloppy terrain where the water is scarce.
 - The method is ideal for row crops, orchard and vineyards.



3. **a. Write manning's uniform flow equation used for canal design.**
b. Design a regime channel (using Lacey's equation) for a discharge of 30 m³/sec with silt factor=1.
Assume trapezoidal section having side slopes 1/2:1

Answer:

a. Manning's formula for canal design is given by:

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

Since,

$$Q = AV$$

Substitute the Manning velocity:

$$Q = \frac{A}{n} R^{2/3} S^{1/2}$$

Where:

Q = Discharge (m³/s)

A = Cross-sectional area (m²)

R = Hydraulic radius (m)

S = Bed slope

n = Manning's roughness coefficient

Velocity equation:

$$V = \frac{1}{n} R^{2/3} S^{1/2}$$

Typical values:

- Smooth concrete canal: 0.011–0.015
- Brick masonry: 0.015–0.017
- Earth canal (good condition): 0.020–0.030
- Rough natural channel: 0.030–0.050 or more

From Manning's equation,

$$V \propto \frac{1}{n}$$

Thus, velocity and discharge are inversely proportional to Manning's roughness coefficient.

The increase in the value of n, results in rough canal surface which reduces the flow velocity and canal discharge. Also, sediment deposition increases due to lower velocity.

The reduction in the value of n, results in smooth canal surface which increases flow velocity and canal discharge. Excessively high velocity may also cause scouring and erosion in unlined canal.

b. Given,

Discharge, Q = 30 m³/s

Silt factor, f = 1

Side slope = 1/2: 1

Therefore, m = 0.5

According to Lacey's equation,

$$V = \left(\frac{Qf^2}{140} \right)^{1/6}$$

For f=1,

$$\begin{aligned} V &= \left(\frac{30}{140} \right)^{1/6} \\ &= 0.774 \text{ m/s} \end{aligned}$$

From, Q=AV

$$\text{Therefore, } A = \frac{Q}{V}$$

$$\begin{aligned} &= \frac{30}{0.774} \\ &= 38.78 \text{ m}^2 \end{aligned}$$

According to Lacey's equation,

$$P = 4.75 \sqrt{Q}$$

$$= 4.75 \sqrt{30} = 26.02 \text{ m}$$

For a trapezoidal channel with side slope 0.5: 1,

$$\text{Area, } A = By + my^2$$

Therefore,

$$38.78 = By + 0.5y^2$$

Wetted perimeter,

$$P = B + 2y \sqrt{1+m^2}$$

Therefore,

$$26.02 = B + 2y \sqrt{1+0.5^2}$$

$$26.02 = B + 2.236y$$

$$B = 26.02 - 2.236y$$

Substituting this value of B in the area equation:

$$38.78 = (26.02 - 2.236y)y + 0.5y^2$$

$$38.78 = 26.02y - 1.736y^2$$

Solving, $y = 1.68 \text{ m}$

$$B = 26.02 - 2.236(1.68)$$

$$= 22.26 \text{ m}$$

$$R = \frac{A}{P}$$

$$= \frac{38.78}{26.02}$$

$$= 1.49 \text{ m}$$

According to Lacey's equation,

$$S = \frac{f^{5/3}}{3340 Q^{1/6}}$$

For $f = 1$ and $Q = 30 \text{ m}^3/\text{s}$,

$$S = \frac{1^{5/3}}{3340 (30)^{1/6}} = 0.00017 = 1: 5887$$

4. What is meant by scour? What precautions do you take against it in weir design? Explain.

Answer:

The erosion of soil and sediment from the river bed or banks due to the high velocity and erosive action of flowing water. In a weir, scour occurs particularly downstream of the weir, where water flows with high velocity and creates turbulence and hydraulic jumps.

The following precautions are taken to protect a weir against scour:

- i. A sufficiently long impervious floor is provided upstream and downstream of the weir to prevent seepage and protect the foundation from erosion.

- ii. Launching apron is provided downstream. It protects the river bed against scour and launches into the scour hole if erosion occurs.
- iii. Downstream floor should be sufficiently long to safely dissipate the energy of flowing water and prevent excessive erosion.
- iv. Hydraulic jump arrangements, stilling basins, baffle blocks, friction blocks and end sills may be provided to reduce the velocity and energy of water.
- v. Cutoff walls are provided at suitable locations, particularly at the upstream and downstream ends of the floor, to increase the seepage path and prevent undermining of the structure.

5. **Write short notes on:**

a. exit gradient

b. Hydraulic jump

c. Retrogression

d. Water measuring devices used in water management.

Answer:

a. Exit gradient:

The hydraulic gradient of seepage flow at the downstream end of a hydraulic structure where seepage water emerges from beneath the impervious floor is known as exit gradient. It is the rate of loss of hydraulic head per unit length at the point where seepage exits. The exit gradient is a criterion for checking the safety of hydraulic structures against piping and undermining.

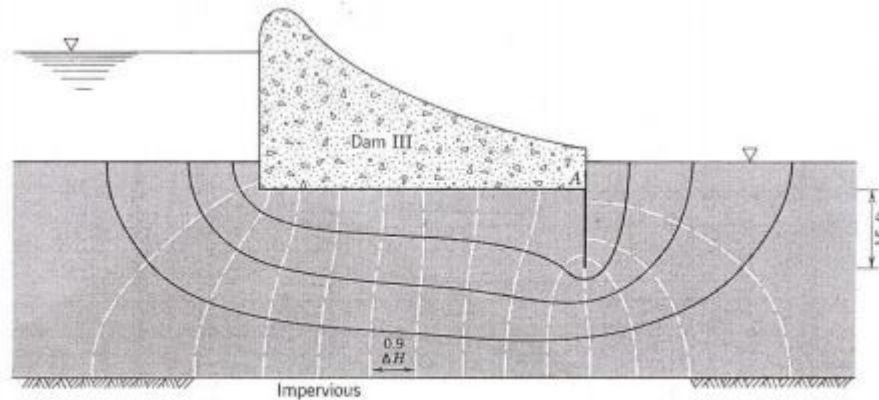


Fig. E18.3 Flow under three dams.

The exit gradient is represented as:

$$G_E = \frac{\Delta h}{\Delta l}$$

where :

$$G_E = \text{Exit gradient}$$

$$\Delta h = \text{Loss of hydraulic head near the exit (m)}$$

$$\Delta l = \text{Length - which the head is lost (m)}$$

If the actual exit gradient exceeds the critical exit gradient, the seepage force becomes greater than the submerged weight of the soil particles, which cause them to move. If the exit gradient is not balanced, structure fails by:

- i. Boiling (Quick Condition)
 - Soil particles lose contact with each other where the soil behaves like a liquid.
- ii. Piping

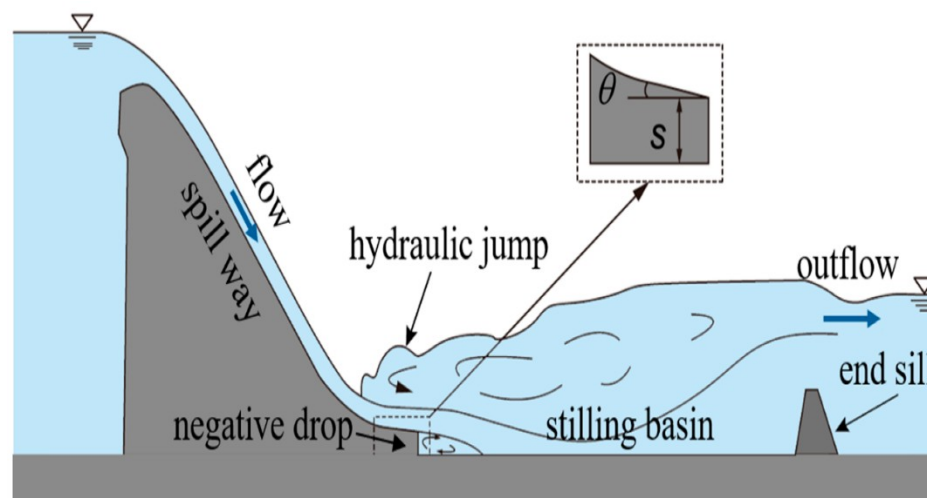
- Fine soil particles are carried away by seepage and underground tunnels (pipes) are formed.
- iii. Undermining of Foundation
 - Continuous removal of soil creates cavities beneath the floor such that the foundation loses support.
- iv. Structural Failure
 - The floor may crack, settle, or heave due to uplift and loss of support which ultimately cause the hydraulic structure to collapse.

b. Hydraulic jump

- When the depth of flow changes rapidly from a low stage to a high stage, there is abrupt rise of water surface. This local phenomenon is known as hydraulic jump.
- It occurs when the flow changes from supercritical to subcritical.
- Hydraulic jump converts kinetic energy of stream flowing rapidly into potential energy.
- It occurs frequently in a channel below a regulating sluice, at toe of spillway and when the channel bed slope suddenly changes from the steeper to flat slope or where the velocity is high.
- Hydraulic jump forms eddy turbulences in water which causes dissipation of energy.

It is necessary to be formed in hydraulic structures which is explained in the points below:

- i. For the dissipation of excess energy of flowing water over spillway and other hydraulic structure.
- ii. For preventing from possible erosion of downstream floor at structure.
- iii. For increasing the weight of apron floor of hydraulic structure due to increased flow depth.
- iv. For mixing chemicals in water by the help of turbulence created.
- v. For raising the water level in the channel for irrigation.
- vi. For increasing the discharge through a sluice by holding back tail water.



For a stable and well-balanced hydraulic jump, the Froude number (Fr) before the jump should be greater than 1 i.e., the flow must be supercritical.

The table below shows the Froude number for different type of hydraulic jump.

Froude Number (Fr_1)	Type of Hydraulic Jump
$Fr = 1$	No hydraulic jump (critical flow)
$(1 < Fr_1 \leq 1.7)$	Undular jump (weak, unstable)
$(1.7 < Fr_1 \leq 2.5)$	Weak jump
$(2.5 < Fr_1 \leq 4.5)$	Oscillating jump
$(4.5 < Fr_1 \leq 9)$	Stable (steady) hydraulic jump
$(Fr_1 > 9)$	Strong jump

A stable and well-balanced hydraulic jump is generally obtained when:

$$4.5 < Fr_1 \leq 9.0$$

c. Retrogression:

Retrogression is the phenomenon in which the river bed level gradually rises due to excessive deposition of sediment, resulting in a reduction in the carrying capacity of the river channel. In river engineering, retrogression may occur when the sediment load entering a river is greater than its sediment-carrying capacity. It causes:

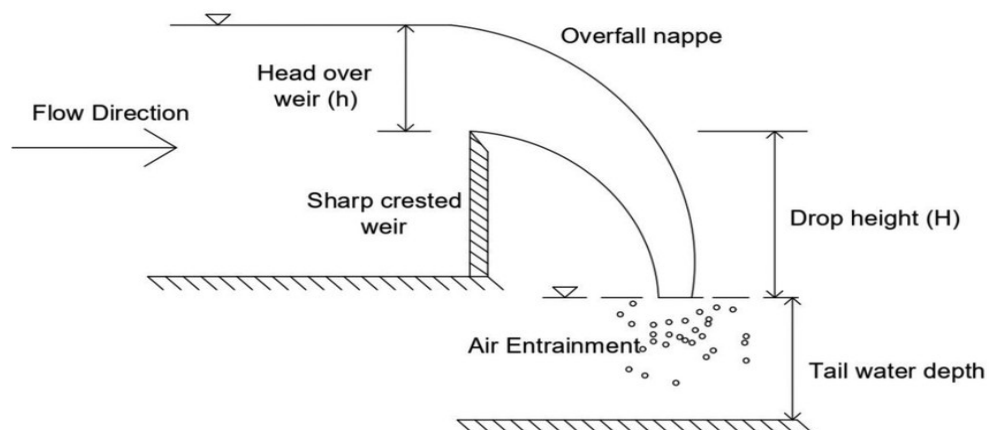
- Raising of the river bed level.
- Reduction in the flood-carrying capacity of the channel.
- Increased frequency of flooding.
- Aggradation and deposition of sediment near hydraulic structures.
- Changes in river channel alignment and morphology.

d. Water measuring device used in water management:

Water measuring devices are used to measure the discharge and volume of water in canals and other structures in irrigation, hydropower and water supply systems for efficient water distribution and management. Some of the water measuring devices used in water management are described below:

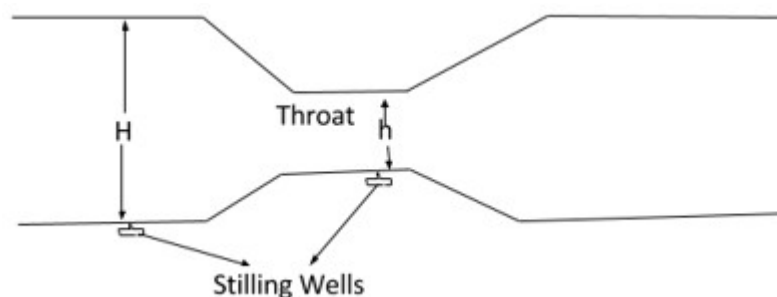
a. Weirs:

- A weir is an impervious barrier constructed across a river or stream to raise the upstream water level, thereby facilitating the diversion of a specific quantity of water into a conveyance system like an irrigation canal or a hydropower intake.
- They are suitable for small to medium discharges.
- There are different types of weirs. i.e., rectangular weir, trapezoidal weir, triangular weirs and so on.



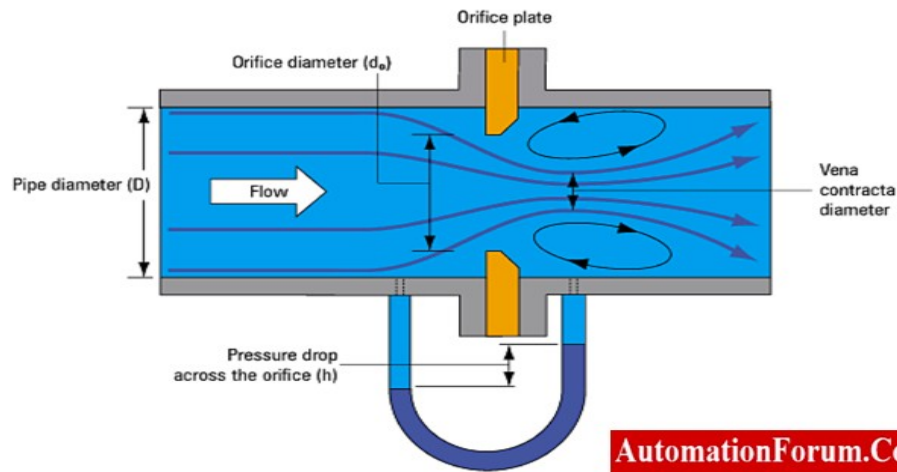
b. Flumes:

- Flumes are suitable for measuring discharges in irrigation canals with minimal head loss.
- Types: Parshall flumes, Cut-throat flume, venturi flume etc.



c. Orifices:

- Orifices are used to measure discharge from reservoir, tanks and canals.
- It is a type of flow meter that measures the flow rate of fluids (liquids or gases) based on the principle of differential pressure.



d. Current meters:

- Measure the velocity of flowing water.
- Discharge is calculated using:

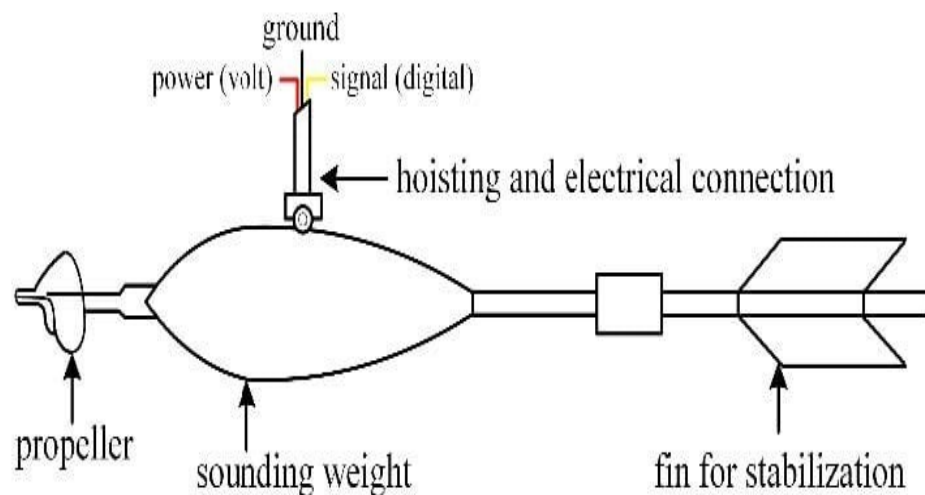
$$Q = A \times V$$

where:

Q = discharge (m^3/s)

A = cross-sectional area (m^2)

V = average velocity (m/s)



6. What are the various stages of river? Why river training is necessary? Describe briefly the different methods adopted in Nepal.

What is the purpose of providing spur in river training in Nepal?

Answer:

There are generally three phases of river. They are:

- i. Youthful stage:
This represents the upper course of river, found in mountainous and hilly regions.
The river at this stage flows at high velocity through the steep slope.
- ii. Mature stage:
This represents the middle course of river, where the slope becomes gentler.
The river valley becomes wider and meanders starts developing.
- iii. Old stage:
This is the lower course of river.
The river at this stage flows at low velocity and has a very gentle slope.

River training works are the structural engineering measures constructed along rivers to control the flow, stabilize the channel, and prevent bank erosion. They are designed to guide water within a specific path, safely pass floodwaters, and protect nearby infrastructure from shifting currents.

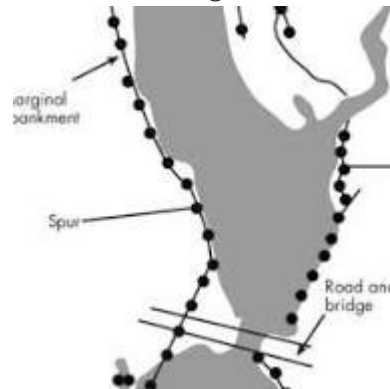
River training works is necessary because:

- i. They protect river banks from erosion by preventing scouring and collapse of river banks.
- ii. They also protect hydraulic structures and ensures safety of bridges, dams, weirs and other structures from erosion and scour.
- iii. They provide safe waterway for navigation by maintaining sufficient depth and stable channel.
- iv. River training works also reduce flood damage by regulating the river flow.
- v. They minimize silt accumulation near bridges, intakes etc.

There are different methods of river training works. Some of them are described below:

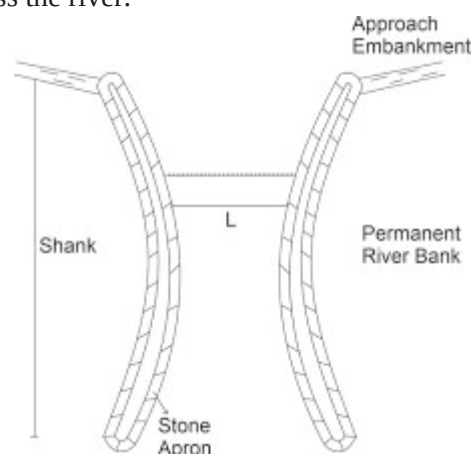
i. Marginal embankments/ levees:

- Marginal embankments are the earthen embankments constructed parallel to the riverbanks to confine flood waters within the cross section available between them.
- They are usually placed at some distance away from the riverbank so that area of the flow is increased.
- Generally required for meandering rivers.



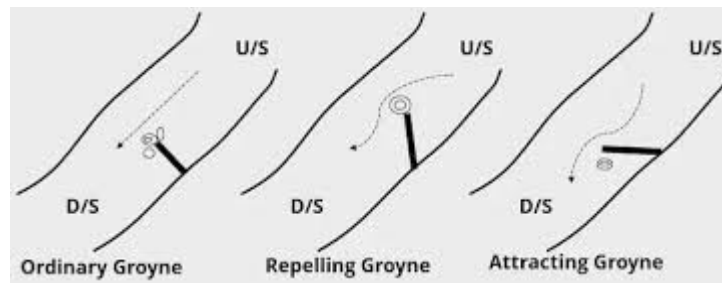
ii. Guide banks:

- Guide bank are the earthen embankments provided on one or both side of river at the bridge site to ensure that the flow path does not change through the waterway at bridge site.
- The layout of guide banks should be such as to guide the flood smoothly through a work constructed across the river.



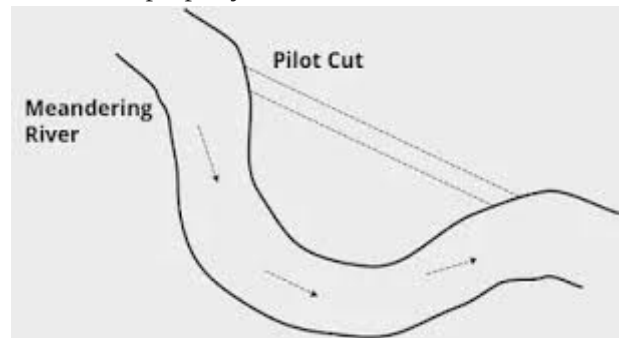
iii. Groynes or spurs:

- Groynes are the structure constructed transverse to the river flow and extend from the bank into the river.
- They help in training the river along a desired course by attracting, deflecting or repelling the flow in the channel.



iv. Artificial cutoff:

- An artificial cutoff is developed in a meandering river to divert the flow along a straight course.
- They are constructed for training of river when its mender goes on increasing and endanger some valuable land or property on the banks.



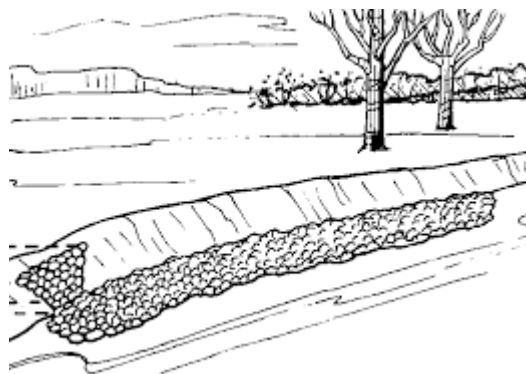
v. Bank protection and Pitched banks:

Bank protection helps in training of the river for the protection of hydraulic structures and adjacent land or property.

Bank protection work can be classified as direct or indirect.

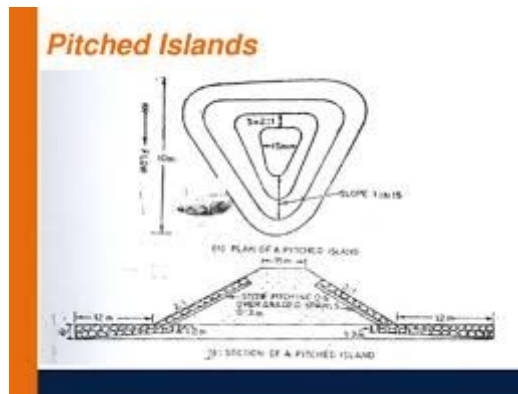
Direct protection works include work done on the bank itself such as vegetation cover, pitching

Indirect protection includes work constructed not directly on the banks but in front of them.



vi. Pitched islands:

- Pitch Island is an artificially created island in the river bed, either made up of masonry or stone pitched on all sides.
- The main purpose of pitched island is to train the river to have an axial flow and improve the channel for navigation purpose.



vii. Bandalling:

- Bandalling is a method of river training works which is used to confine the low flow water of a shallow river in a single channel for maintaining the required depth for navigation.
- Bandalling started after the flood season is over and when the water level begins to fall.



The purpose of providing spur in river training in Nepal are listed below:

- i. Spurs protect the river banks by deflecting the main current away from the vulnerable river bank.
- ii. They guide the flow of river along the desired alignment and prevent excessive lateral shifting of river.
- iii. They also create a relatively calm zone near the bank reducing the erosive forces.
- iv. They confine the river to a relatively stable channel and reduce unwanted meandering.
- v. Properly designed spurs also redirect the high-velocity flood flows away from vulnerable areas.

7. What is water logging? What are its effects? Explain various methods of reclamation of water-logged areas?

Answer:

Water logging is the condition of saturation of soil such that air phase is restricted and anaerobic conditions prevails. Waterlogging is often accompanied by soil salinity as waterlogged soils prevent leaching of the salts imported by the irrigation water. Due to water logging, groundwater table rises close to the ground surface (generally within 1.5–2.0 m), causing the root zone to remain saturated and reducing soil aeration.

The effects of water-logging are listed below:

i. Reduced Soil Fertility and Nitrification:

Water logging causes lack of oxygen in saturated pores hinders the process of nitrification, which significantly reduces crop yields.

ii. Accumulation of Toxic Substances:

Due to the absence of oxygen, anaerobic decomposition occurs, leading to the formation and buildup of toxic substances such as hydrogen sulphide, methane, and ferrous compounds.

iii. **Soil Salinity:**

Waterlogging prevents the leaching of salts. High groundwater levels bring these dissolved salts to the surface through capillary action, resulting in a high salt concentration in the surface layer that is harmful to plants.

iv. **Nutrient Deficiency:**

It causes essential nutrient imbalances, particularly the loss of nitrogen, which is critical for plant growth.

v. **Reduced Soil Temperature:**

Waterlogged soils warm up more slowly than well-drained soils, resulting in lower soil temperatures that can further retard plant growth.

The various methods of reclamation of water-logged areas are as follows:

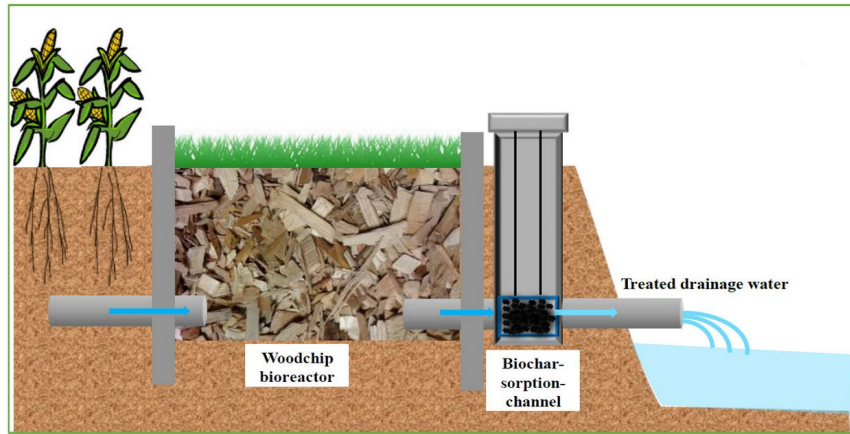
i. **Surface drainage system:**

- The removal of excess rainwater falling on the fields or excess irrigation water applied to the fields by constructing open ditches. Field drains and other structures is known as surface drainage system.
- The land is loped toward the ditches or drains making excess flow of water to remove from the drains.
- Surface drains can be divided into surface drain and deep surface drains.



ii. **Subsurface drainage system:**

- The removal of excess rainwater by using buried perforated pipes or tile drains to lower the groundwater table is known as subsurface drainage system.
- It is most effective for controlling water logging in agricultural fields.



iii. **Interceptor drains:**

- These drains are constructed along the canals or hill slope to intercept seepage before it reaches agricultural land.

iv. **Bio-drainage:**

- Plantation of deeply rooted trees, highly water absorbing plants to lower the groundwater table naturally.

v. **Open drainage system:**

- It consists of open ditches that carry water to nearby rivers or canals.
- It is simple and economical for large agricultural areas.

8. **Define confined and unconfined aquifer. If an artesian well produces 250 litre per minute with a drawdown of 3 m in the pumping well, what will be the discharge with 4 m drawdown.**

Answer:

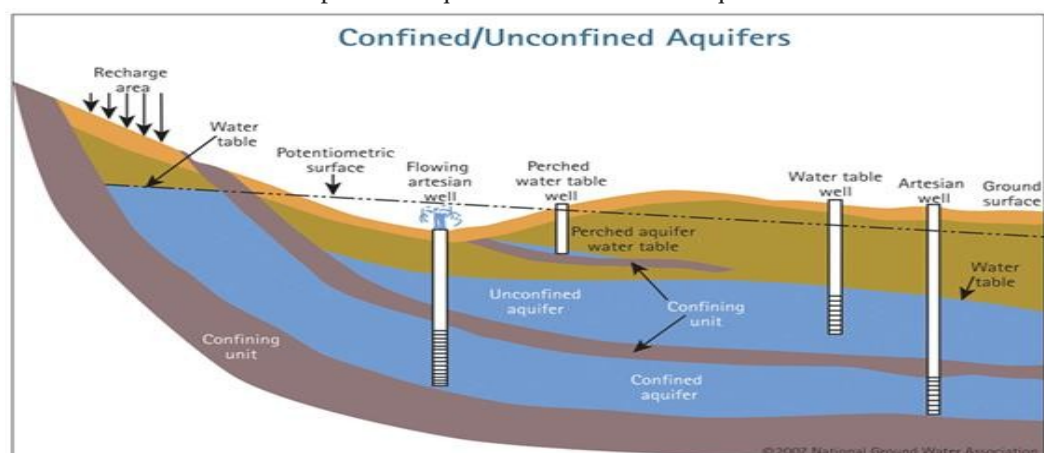
- An aquifer may be defined as the geological formation that contains sufficient permeable material that permits storage as well as transmission of water through it under ordinary conditions.
- Water can be withdrawn from aquifer for example: sand layer.
- Aquifer are of two types:

i. **Confined aquifer:**

- It is the type of aquifer in which ground water is confined under pressure greater than atmospheric pressure by overlying relatively impermeable strata.
- Confined aquifer generally occurs at significant depth below the ground surface.
- It is known as artesian aquifer or pressure aquifer.

ii. **Unconfined aquifer:**

- It is the type of aquifer in which water table forms the upper surface of zone of saturation.
- It is also known as phreatic aquifer or non-artesian aquifer.



Given,

Initial discharge, $Q_1 = 250$ litres/minute

Initial drawdown, $s_1 = 3$ m

New drawdown, $s_2 = 4$ m

For an artesian well under steady-state conditions, discharge is directly proportional to drawdown.

Therefore,

$$\frac{Q_1}{s_1} = \frac{Q_2}{s_2}$$

Hence,

$$Q_2 = Q_1 \times \frac{s_2}{s_1}$$

Substituting the given values,

$$Q_2 = 250 \times \frac{4}{3}$$

$$Q_2 = 333.33 \text{ litres / minute}$$

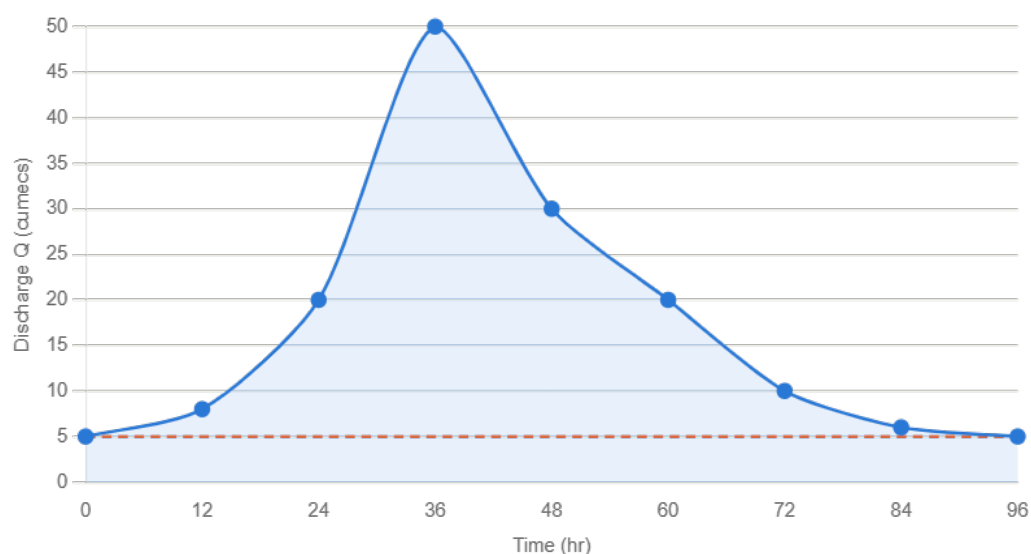
9. The following stream flow records are obtained from a gauging station.

Time (hr)	0	12	24	36	48	60	72	84	96
Q (cumecs)	5	8	20	50	30	20	10	6	5

Determine the volume of flood runoff, base flow, surface runoff and peak flood.

Answer:

From the hydrograph, the discharge before and after the flood is $5 \text{ m}^3/\text{s}$. Therefore, the base flow is assumed to be $5 \text{ m}^3/\text{s}$ throughout the flood period.



Base flow discharge, $Q_b = 5 \text{ m}^3/\text{s}$

Duration of flood, $T = 96$ hr

Therefore,

$$V_b = Q_b \times T$$

$$= 5 \times 96 \times 60 \times 60$$

$$= 1728000 \text{ m}^3$$

$$\text{Volume of the base flow} = 1728000 \text{ m}^3$$

Total flood runoff volume:

Using trapezoidal method,

$$V = \frac{Q_1 + Q_2}{2} \times \Delta t$$

$$\Delta t = 12 \times 60 \times 60$$

$$\Delta t = 43,200 \text{ s}$$

Therefore,

$$V_f = \left[\frac{5+8}{2} + \frac{8+20}{2} + \frac{20+50}{2} + \frac{50+30}{2} + \frac{30+20}{2} + \frac{20+10}{2} + \frac{10+6}{2} + \frac{6+5}{2} \right] \times 43200$$

$$= [6.5 + 14 + 35 + 40 + 25 + 15 + 8 + 5.5] \times 43200$$

$$= 149 \times 43200$$

$$= 6436800 \text{ m}^3$$

$$\text{Total flood runoff volume} = 6436800 \text{ m}^3.$$

Surface Runoff volume:

$$V_s = V_f - V_b$$

$$= 6436800 - 1728000$$

$$= 4708800 \text{ m}^3.$$

$$\text{Peak flood, } Q_{\max} = 50 \text{ m}^3/\text{sec}.$$

10. Define “infiltration” of rain water and “infiltration capacity” of the area. Describe the factors affecting infiltration capacity.

Answer:

The process by which water on the ground surface enters and moves into the soil through its pores is known as infiltration of rain water. It is an important part of the hydrologic cycle because it determines how much rainfall becomes groundwater recharge and how much becomes surface runoff.

The rate at which water can enter the soil surface under given conditions over the specified area is known as infiltration capacity of that area. It depends on the properties of the soil, surface conditions, vegetation, rainfall and other factors.

The factors affecting the infiltration capacity are as follows:

- i. Soil texture:
 - Soil texture affects the infiltration rate.
 - Sandy soil has larger pores and has high infiltration capacity.
 - Clayey soil has smaller pores and has lower infiltration capacity.
- ii. Soil structures:
 - Well-graded and granular soils have interconnected pores that allow water to infiltrate easily.
 - Compacted or poorly structured soils have lower infiltration capacity.

- iii. Porosity of soil:
 - Higher porosity provides more pore spaces for water movement and increases infiltration capacity.
- iv. Compaction of soil:
 - Compaction reduces pore spaces and permeability.
 - Hence, adequately compacted soil generally has lower infiltration capacity.
- v. Vegetation and land use:
 - Vegetation increases the infiltration capacity by reducing surface runoff and increasing soil porosity through roots.
 - Deforestation and settlement reduce the infiltration capacity of the soil.
 - Hence, urban areas are experiencing floods in settlement areas due to excess surface runoff and lack of infiltration.
- vi. Rainfall intensity:
 - When rainfall intensity is less than infiltration capacity, almost all rainfall can infiltrate into the soil, so infiltration rate equals rainfall intensity and there is little or no surface runoff.
 - When rainfall intensity exceeds infiltration capacity, the soil cannot absorb water as quickly as it is supplied. Once the soil surface becomes saturated, the actual infiltration rate becomes approximately equal to the infiltration capacity, while the excess rainfall forms surface runoff.